

EXECUTIVE SUMMARY

Being findable and being citable are now different problems

Ranking #1 on a search results page used to be enough, a human would scroll to your link. An answer engine skips the page entirely: it retrieves a handful of passages, reasons over them, and writes one answer. Your brand either enters that working set and gets named, or it doesn't exist for that query.

Sections 1–3 define AI visibility as a measurable quantity. Section 4 models it as a logistic function of six evidence factors and *proves* the function is strictly increasing in each one (Theorem 1), worked through with a running numeric example, a brand called Acme, down to the 13 raw metrics (backlinks, referring domains, review volume, and so on) behind its scores. Section 5 then asks the practical question, given a limited budget, which lever to fix *first*, and proves it's the cheapest high-coefficient one, not the biggest coefficient. Section 6 adds a game-theoretic argument, a signaling model, for why independent evidence is trustworthy in the first place, and why buying it (fake reviews, link farms) doesn't work. Section 7 turns all of this from theory into a measurement protocol, including a control-group design for isolating causal effect from background noise.

Core objective

$$\max_B \mathbb{E}_{q \sim Q} [P_B(q)]$$

Maximize the expected probability that the AI includes brand B across the query universe Q a customer actually asks.

A brand either appears in the answer, or it doesn't

Fix a brand B and a finite set of commercially relevant questions, the *query universe*:

$$Q = \{q_1, q_2, \dots, q_n\}$$

Worked example, used throughout this paper (an illustrative composite, not an actual company; see Section 10). Let $B = \text{Acme}$, a project-management SaaS for creative agencies. A realistic Q might read: "best project management software for agencies," "alternatives to Asana," "how should an agency manage client approvals?" and "Acme vs Asana", the informational, commercial, comparative and brand questions a real buyer asks before purchasing. Define the appearance indicator:

$$V(B, q) = \begin{cases} 1 & \text{if } B \text{ appears in the AI answer to } q \\ 0 & \text{otherwise} \end{cases}$$

The simplest visibility statistic is the sample mean of this indicator over the query universe:

DEFINITION 1: AI VISIBILITY RATE

$$\text{AIV}(B) = \frac{1}{n} \sum_{i=1}^n V(B, q_i)$$

If B appears in 37 of 100 representative queries, $\text{AIV}(B) = 0.37$. This is already a measurable KPI, Section 7 treats it as a statistical estimator with a confidence interval, not just a percentage.

Mentioned, recommended, and cited are not the same event

Refine the single indicator V into three binary sub-indicators of increasing commercial value:

$$M(B, q), R_{\text{ec}}(B, q), C(B, q) \in \{0, 1\}$$

these are mentioned, recommended, and cited (backed by a named source), respectively, and combine them into a weighted score:

$$S(B, q) = w_m M(B, q) + w_r R_{\text{ec}}(B, q) + w_c C(B, q), \quad w_m + w_r + w_c = 1$$

With, say, $w_m = 0.2$, $w_r = 0.4$, $w_c = 0.4$: a bare mention scores 0.2; a recommendation scores 0.4; a recommendation backed by a citation scores 0.8. This is a considerably more useful signal than a binary "we showed up", it separates being named from being endorsed. If Acme is recommended for "best project management software for agencies" and the answer cites a G2 review page, $S(\text{Acme}, q) = 0.4 + 0.4 = 0.8$; a bare mention with no endorsement scores only 0.2.

3. THE OBJECTIVE AS AN EXPECTATION

Treat appearance as a probability, not a binary fact

A given brand doesn't deterministically appear or not for a given query, sampling temperature, retrieval variance, and index drift make it a random event. Define:

$$P_B(q) := \Pr(B \text{ appears in the answer} \mid q)$$

and the optimization target is the expectation of this probability over the query universe a real buyer asks:

THE OBJECTIVE

$$\max_B \mathbb{E}_{q \sim Q} [P_B(q)]$$

In plain language: maximize the probability that the AI includes your brand, averaged over the commercially relevant questions people actually ask. Section 4 asks what $P_B(q)$ is a function of.

4. THE EVIDENCE MODEL

Theorem 1: citation probability is monotonic in every lever

No AI search provider publishes its ranking function, and this is not a claim to have reverse engineered one. It is a stylized, testable model, a logistic function of six factors known to matter to retrieval-and-generation systems generally, built so its qualitative predictions (monotonicity, diminishing returns) hold for a wide family of coefficient values, not one guessed set of weights.

DEFINITION 2: THE SIX EVIDENCE FACTORS

Let $R, A, E, T, S, F \in [0, 1]$ denote, respectively, a brand's relevance to the query, authority, independent evidence, topical coverage, semantic/entity clarity, and content freshness. Define the linear predictor and the logistic citation probability:

$$z_B(q) = \beta_0 + \beta_1 R + \beta_2 A + \beta_3 E + \beta_4 T + \beta_5 S + \beta_6 F$$
$$P_B(q) = \sigma(z_B(q)), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

THEOREM 1: MONOTONICITY

If $\beta_k > 0$ for every $k = 1, \dots, 6$, i.e. each factor genuinely helps and none hurts, then for every factor $x_k \in \{R, A, E, T, S, F\}$:

$$\frac{\partial P_B}{\partial x_k} = \beta_k \sigma(z_B) (1 - \sigma(z_B)) > 0$$

Proof. The logistic function satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, which is strictly positive for every finite z since $\sigma(z) \in (0, 1)$. By the chain rule, $\partial P_B / \partial x_k = \beta_k \cdot \sigma'(z_B)$. Since $\beta_k > 0$ by hypothesis and $\sigma'(z_B) > 0$ always, the product is strictly positive. ■

Every lever that genuinely carries positive weight strictly raises citation probability, none of the six factors can be safely ignored, and none can hurt you by being improved.

COROLLARY 1: DIMINISHING RETURNS, AND WHY TO DIVERSIFY

Differentiating again, $\sigma''(z) = \sigma'(z)(1 - 2\sigma(z))$, which changes sign at $\sigma(z) = \frac{1}{2}$ (i.e. $z = 0$). For $z > 0$, $\sigma''(z) < 0$: the marginal gain $\sigma'(z)$ is *falling* as z grows. Since $\partial^2 P_B / \partial x_k^2 = \beta_k^2 \sigma''(z_B)$ has the same sign as $\sigma''(z_B)$, P_B is concave in each x_k once $z_B > 0$.

$$\frac{\partial^2 P_B}{\partial x_k^2} = \beta_k^2 \sigma'(z_B)(1 - 2\sigma(z_B)) < 0 \quad \text{once } z_B > 0$$

A budget-constrained optimum equalizes marginal effect $\beta_k \sigma'(z_B)$ across the six levers rather than exhausting it on whichever is already strongest. In practice: once one lever (say authority) has already pushed z_B well past zero, another unit of authority buys less citation probability than the same unit spent raising a neglected lever (say freshness), the formal reason a diversified GEO program outperforms a single-tactic one.

WORKED EXAMPLE: ACME'S BASELINE

Suppose a fitted (illustrative) coefficient set is $\beta_0 = -3.8$, and $(\beta_1, \dots, \beta_6) = (1.0, 1.5, 1.5, 0.8, 1.2, 0.5)$ for (R, A, E, T, S, F) . Acme's audit today, before any intervention, scores:

Factor	R	A	E	T	S	F
Acme's score	0.50	0.15	0.20	0.45	0.10	0.60

Decent content (R, F) but a weak, inconsistent entity presence ($S = 0.10$): "Acme," "Acme Software" and "Acme Inc." all appear across its own site, LinkedIn and G2. Plugging in:

$$z_B = -3.8 + (1.0)(0.50) + (1.5)(0.15) + (1.5)(0.20) + (0.8)(0.45) + (1.2)(0.10) + (0.5)(0.60) \\ P_B(q) = \sigma(-1.995) \approx 0.120$$

A predicted 12% baseline citation probability, matching the $\hat{p}_{\text{before}} = 12\%$ measured for Acme's optimized-topic group in Section 7's difference-in-differences example. Section 5 asks: of the six levers, which is the most efficient one to fix first?

ACME'S RAW AUDIT SHEET: FROM 13 METRICS TO SIX FACTORS

The R, A, E, T, S, F scores above aren't measured directly, they're a normalized roll-up of the ordinary metrics an audit tool or analyst actually pulls. Below is the audit behind Acme's baseline, each raw metric mapped to the factor it feeds, benchmarked against the category leader:

Raw metric	Feeds	Acme	Category leader
Referring domains	A	85	~600
Backlinks (total)	A	340	~12,000
Domain authority (0–100)	A	28	~74
Independent mentions (12mo)	E	6	~140
High-authority citations	E	1	~18
Review volume (G2+Capterra+Trustpilot)	E	42	~2,300
Third-party articles (independent)	E	9	~85
Comparison pages ("Acme vs X")	T	3, all self-authored	40+, mostly independent
Page depth (of 30 buyer questions in <i>Q</i> with a dedicated page)	T	14/30	28/30
Topical relevance (avg. embedding similarity to <i>Q</i>)	R	0.52	0.81
Structured-data completeness (of 10 core schema fields)	S	2/10, 3 conflicting name variants	10/10, 1 canonical name
Content freshness (median days since last fact-check)	F	45 days	12 days
Brand-search volume (monthly)	check	1,100	~48,000

Rows 1–12 roll up, after min-max normalization against the category benchmark column, into the *R*, *A*, *E*, *T*, *S*, *F* scores above: *A* is dragged down by thin backlinks and domain authority; *E* by almost no high-authority citations or independent articles; *T* because Acme's three comparison pages are all self-authored, none of the leader's 40+ are; *S* by the split naming and near-empty schema. Brand-search volume isn't one of the six factors, it's the cross-check used in Section 6: a real signal of quality should show up there too.

Where to start, given a limited budget

Theorem 1 says every lever helps. It doesn't say which to fix first. Real levers cost different amounts to move: unifying your own entity data is days of internal work; earning independent press coverage is months of relationship-building you don't fully control. The efficient question isn't "which β_k is biggest," it's "which β_k per dollar (or per week) is biggest."

PROPOSITION 2: THE EFFICIENT FIRST MOVE

Let $c_k > 0$ be the cost of a one-unit improvement to factor x_k , and $\bar{c}$ a fixed budget. Locally (for a small budget, around the current z_B), the first-order gain from increments $\Delta_k \geq 0$ is:

$$\Delta P_B \approx \sigma'(z_B) \sum_k \beta_k \Delta_k, \quad \text{subject to} \quad \sum_k c_k \Delta_k \leq \bar{c}$$

This is a fractional-knapsack allocation problem: $\sigma'(z_B) > 0$ is a common positive scale factor, so maximizing ΔP_B is equivalent to maximizing $\sum_k \beta_k \Delta_k$ subject to the same budget constraint. The classical result for this linear program is that the optimum is a corner solution: spend the entire budget on the single lever with the highest benefit-to-cost ratio.

$$k^* = \arg \max_k \frac{\beta_k}{c_k}, \quad \Delta P_B \approx \sigma'(z_B) \cdot \frac{\beta_{k^*}}{c_{k^*}} \cdot \bar{c}$$

Iterating this (greedily investing in whichever remaining lever has the highest β_k/c_k , then re-ranking as costs rise per Corollary 1) is the standard greedy algorithm for fractional knapsack, provably optimal for the linearized objective at each step.

WORKED EXAMPLE: RANKING ACME'S SIX LEVERS BY EFFICIENCY

Continuing the baseline from Section 4, estimate a rough cost (in person-weeks of effort) for moving each factor meaningfully, and rank by β_k/c_k :

Lever	β_k	Cost c_k (weeks)	β_k/c_k	Rank
S: semantic clarity	1.2	1	1.20	1st
F: freshness	0.5	1	0.50	2nd (tie)

Lever	β_k	Cost c_k (weeks)	β_k/c_k	Rank
R: relevance	1.0	2	0.50	2nd (tie)
T: topical coverage	0.8	3	0.27	4th
E: evidence	1.5	6	0.25	5th
A: authority	1.5	8	0.19	6th (last)

Authority and evidence have the *largest raw* β_k (tied at 1.5), but also the highest cost, months of press and review-earning Acme doesn't fully control. Semantic clarity is cheap (a naming and schema cleanup, roughly a person-week) and has a strong β_k , giving it by far the best ratio. The efficient first move is to fix S , not to chase the biggest coefficient.

Applying Proposition 2, Acme raises S from 0.10 to 0.90 (unify "Acme" everywhere; ship Organization/Product schema) and holds every other factor at its Section 4 baseline:

$$z'_B = -1.995 - (1.2)(0.10) + (1.2)(0.90) = -1.035, \quad P_B(q) = \sigma(-1.035) \approx 0.262$$

A single, low-cost, one-week fix moves the model's predicted citation probability from 12% to roughly 26%, before a single unit of expensive authority or evidence has been earned. This is exactly why the 90-day plan in Section 9 puts S (and the equally cheap F) in weeks 3–5, and saves the slow, expensive A and E for weeks 10–13: it is the greedy fractional-knapsack ordering from the table above, not an arbitrary sequence.

Why one page you wrote is worth less than five independent ones

The authority and evidence factors above (A, E) are themselves built from many independent sources. Model each source i as contributing evidence e_i , weighted by its own credibility w_i , into a pooled evidence score, which crosses a threshold θ before the system is confident enough to recommend:

$$E_B = \sum_{i=1}^k w_i e_i, \quad \Pr(\text{recommend} \mid E_B) = \sigma(E_B - \theta)$$

Acme, pre-intervention, might sit at $E_B = 1.2$; an established competitor at $E_B = 7.8$; the category leader at $E_B = 15.4$, as independent, credible evidence accumulates, $\sigma(E_B - \theta)$ rises toward 1. Treated as a graph, brand, product, category, publication, and customer nodes, connected by "mentions" and "reviews" edges, the practical goal is a *dense, consistent subgraph* around the brand:

MODELING HEURISTIC, NOT A DERIVED IDENTITY

Brand strength $\propto$ Authority $\times$ Relevance $\times$ Evidence density $\times$ Consistency

Unlike Theorem 1, this proportionality is a conceptual heuristic for why distributed third-party evidence compounds rather than adds, it is not proved from first principles here. It explains, informally, why 500 brand-authored blog posts move the needle far less than 20 independent ones: they add to only one node's out-degree, not to the density of the surrounding subgraph.

PROPOSITION 3: EVIDENCE AS A COSTLY SIGNAL (SPENCE, 1973)

The heuristic above says independent evidence compounds. Game theory says why an AI system is *right* to trust it. Model each brand as having a true, unobserved quality $\tau \in \{\tau_L, \tau_H\}$, $\tau_H > \tau_L > 0$, the AI system cannot see directly. It observes only the evidence level E_B built up above, and recommends the brand if it does so, worth v to the brand. Producing E_B isn't free: assume a cost

$$c(E_B, \tau) = \frac{E_B^2}{2\tau}$$

This satisfies the single-crossing (Spence-Mirrlees) condition, marginal cost falls as true quality rises, a genuinely good product earns reviews and press more cheaply than a mediocre one manufacturing the same volume. In the least-cost separating equilibrium, the low type produces nothing ($E_{B,L}^* = 0$) and the high type produces just enough that the low type is indifferent to imitating it:

$$c(E_{B,H}^*, \tau_L) = v \implies \frac{(E_{B,H}^*)^2}{2\tau_L} = v \implies E_{B,H}^* = \sqrt{2v\tau_L}$$

Check. At $E_{B,H}^*$, the low type's cost to mimic is $c(E_{B,H}^*, \tau_L) = v$, exactly the reward, so it's indifferent and (with any tie-break toward truth-telling) reveals itself. The genuine high type's cost is $c(E_{B,H}^*, \tau_H) = v\tau_L/\tau_H < v$, strictly less than the reward, so it strictly prefers to signal. The equilibrium separates, and evidence becomes fully informative: this is the formal reason Theorem 1 is right to give E a positive, trusted β_k .

Corollary 2, why buying fake reviews backfires. The equilibrium only holds while faking E_B is genuinely costly for the low type. Link farms, incentivized reviews and AI-written "independent" roundups all lower that cost; providers respond by investing in detection precisely because a cheaply-faked signal stops separating types and E 's trustworthiness (and effective β_k) collapses toward zero for everyone, not just the cheater. A cheap sanity check: cross E_B against something a low type can't cheaply fake, brand-search volume (row 13 of Section 4's audit sheet). Evidence rising while branded search stays flat is the signature of manufactured, not earned, signal.

From estimator to causal proof

7.1: POINT ESTIMATE AND CONFIDENCE INTERVAL

Sampling n queries and counting x appearances gives the estimator $\hat{p} = x/n$ for $\text{AIV}(B)$, with the usual normal-approximation interval:

$$\hat{p} = \frac{x}{n}, \quad \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

7.2: BEFORE / AFTER LIFT

Continuing the Acme example: with $n = 1,000$ prompts, $x = 120$ mentions before optimization gives $\hat{p}_{\text{before}} = 12\%$, matching Section 4's model prediction; $x = 230$ after gives $\hat{p}_{\text{after}} = 23\%$:

$$\Delta = \hat{p}_{\text{after}} - \hat{p}_{\text{before}} = 11 \text{ pp}, \quad \frac{\Delta}{\hat{p}_{\text{before}}} = \frac{11}{12} \approx 91.7\% \text{ relative lift}$$

Measurable, but not yet *causal*. AI visibility can drift for everyone as models and indices change; a single before/after comparison can't separate your intervention from that drift.

7.3: DIFFERENCE-IN-DIFFERENCES, ISOLATING THE CAUSAL EFFECT

Acme runs the Section 5 playbook (fix S and F , then R and T , then A and E) on 50 topics, and holds out 50 comparable topics as an unoptimized control:

	Before	After
Optimized topics	12%	28%
Control topics	11%	14%

$$\widehat{\text{DiD}} = (\hat{p}_{T,\text{after}} - \hat{p}_{T,\text{before}}) - (\hat{p}_{C,\text{after}} - \hat{p}_{C,\text{before}}) = (28 - 12) - (14 - 11) = \boxed{13 \text{ pp}}$$

The control group's 3-point drift is background noise (new model version, seasonal demand, whatever). Subtracting it leaves 13 points attributable to the intervention, far stronger evidence

than "our AI visibility went up."

7.4: A COMPOSITE TRACKING SCORE AND COMPETITIVE SHARE

For a weekly dashboard, combine visibility, recommendation rate, citation rate, and competitive share of voice into one composite:

$$\text{AIS} = V \times \text{Rec} \times \text{Cit} \times \text{SOV}$$

where $V = \text{AIV}(B)$ from Definition 1, Rec is the share of appearances that are recommendations, Cit is the share that are source-backed, and SOV_B is brand B 's share of all tracked brands' appearances:

$$\text{SOV}_B = \frac{\text{appearances}_B}{\sum_j \text{appearances}_j}$$

e.g. across 100 category questions, if the four tracked brands appear 78, 63, 52 and 19 times respectively, the fourth brand's share of voice is $19/(78 + 63 + 52 + 19) = 19/212 \approx 8.96\%$, a genuinely competitive metric, unlike "we rank #3 for one keyword."

7.5: TESTING WHICH FACTORS ACTUALLY PREDICT VISIBILITY

With a panel of (query, brand, engine, date) observations and covariates $X_1, \dots, X_K$ (referring domains, structured-data completeness, review volume, etc.), fit:

$$\Pr(Y = 1 \mid X) = \sigma\left(\beta_0 + \sum_{k=1}^K \beta_k X_k\right), \quad Y = 1 \iff \text{brand appeared}$$

This estimates *association*, not causation, a positive $\hat{\beta}_k$ says a factor predicts visibility, not that moving it will change your own outcome. The difference-in-differences design in §7.3, run on your own intervention, is what upgrades a correlational finding into causal evidence.

What Theorem 1's factors mean in practice

R

Relevance

How closely your content matches the actual questions people ask, not the keywords you rank for. Answer-first structure and full intent-cluster coverage raise this.

A

Authority

Independent, credible sources treating you as a trustworthy reference in your category, press, analysts, high-authority reviews.

E

Evidence

The same claim about your brand, independently repeated across press, reviews, forums and docs, corroboration, not repetition on your own site.

T

Topical coverage

Breadth across the full query universe Q , informational, commercial, comparative and problem-oriented questions, not just your brand name.

S

Semantic clarity

One canonical name, category and description everywhere you appear. Every naming variant adds disambiguation risk the model must resolve before it will name you.

F

Freshness

Facts (pricing, specs, positioning) kept current. Stale, contradicted claims lower a system's confidence in citing you at all.

9. THE PLAYBOOK

Turning Theorem 1 into a 90-day plan

WEEKS 1–2: BASELINE

Build your query universe Q (30+ real buyer questions) and measure $\hat{p}_{\text{before}}$ across ChatGPT, Gemini and AI Overviews. Hold out a comparable control set of topics per §7.3.

WEEKS 3–5: S & F FIRST

Fix semantic clarity and freshness first, the highest benefit-per-cost levers per Section 5's Proposition 2. Unify naming; ship Organization/Product schema; correct stale facts.

WEEKS 6–9: R & T

Raise relevance and topical coverage: rewrite key pages answer-first, one direct claim per heading, covering the full intent cluster in Q rather than the head keyword alone.

WEEKS 10–13: A & E

Build authority and evidence: earn independent corroboration of the same core claims across press, reviews and forums. Re-run $\hat{p}$ on both treatment and control, and compute $\widehat{\text{DiD}}$.

What this model is, and isn't

- **Acme is an illustrative composite, not an actual company.** Every number attached to it, its factor scores, coefficients, costs, the 12%/26%/28% figures in Sections 4, 5 and 7, and the 13-row audit sheet in Section 4 (backlinks, referring domains, review counts, and the rest), is a worked example chosen to be internally consistent and realistic in scale, not a measurement of any real business.
- $\sigma(z_B)$ in Section 4 is a stylized model built to be qualitatively robust, not a disclosed or reverse-engineered ranking function of any specific AI search product, those are proprietary and change frequently.
- Theorem 1 and Corollary 1 are proved *from the model's stated assumptions* ($\beta_k > 0$), which is a directional claim, "these factors don't hurt, and returns diminish", not a numeric guarantee for any one brand. Proposition 2's costs c_k are illustrative planning estimates, not measured figures either.
- Section 6's proportionality heuristic and Proposition 3's signaling model are both stylized: real evidence-production costs aren't literally quadratic, and no AI provider is known to run a two-type Spence game internally. The result that matters is qualitative, corroborated evidence is credible because and only while it stays costly to fake, not the specific functional form.
- Section 7.5's regression establishes association; only the controlled experiment in §7.3, run on your own site, supports a causal claim about your own intervention.
- Be skeptical of anyone selling a specific "AI schema" as a guaranteed unlock, the model here argues for a portfolio of ordinary, verifiable signals (relevance, authority, evidence, coverage, clarity, freshness), not a trick.